

Internet of Things

The name given to the network of interconnected devices with embedded computers.

interpreter

A program that translates and executes the source code of another computer program one line at a time.

key

In cryptography, any additional information that is used to encrypt or decrypt a message.

keylogger

A computer malware program that tracks what keyboard keys are pressed in order to gain access to passwords and other information.

logic gate

A device that carries out computer calculations with binary numbers.

machine code

Instructions that a computer can understand and uses to send instructions to the computer's hardware.

machine learning

The branch of computer science that gives a computer the ability to learn to do something with data, rather than explicitly programming it.

malware

Malicious software that gains illegal access to a computer or system. Malware includes worms, viruses, spyware,

trojans, rootkits, ransomware, backdoors, and hybrid threats.

microprocessor

Another name for the central processing unit (CPU).

motherboard

The main circuit board of a computer, which holds the CPU, memory, and connectors for the hard drive and optical drives, among other connections. The motherboard allows hardware components of a computer to communicate with each other.

net neutrality

The principle that governments or ISPs should not block, discriminate, or charge users differently to access the internet.

network

A group of connected devices that can share resources and data. Networks can be classified by size or topology (layout).

node

Any device that sends or receives data through a network.

object-oriented language

A programming language that includes the concept of objects that model real-world things. An object usually has fields (containing data) and methods (containing code) that represent behaviours.

online identity

The self or selves that a user presents to the online world.

operating system (OS)

A piece of software that manages a computer's hardware and software resources, and makes it easier for them to be used. The most common examples are Linux, macOS, and Windows.

output

Something that is produced by a person or machine, or the place where information leaves a system.

pattern recognition

The process of seeing repetition in a predictable manner. Pattern recognition is an important part of computational thinking.

peripheral

Any piece of hardware that allows users to interact with a computer.

photoresist

A substance that is used in the manufacture of computer chips that protects parts of a chip in development from being eroded by the manufacturing process.

pixel

Short for picture element. A tiny area of illumination on a computer display screen, one of many from which an image is composed.

programming

The process of giving instructions to a computer.

programming language

A formalized set of words and symbols that allows a

person to give instructions to a computer. High-level languages are closer to human language. Low-level languages are closer to binary code.

program

A collection of instructions that performs a specific task when executed by a computer.

protocol

A set of rules that governs the transmission of data between devices; for example, Hypertext Transfer Protocol, which is used for visiting websites.

random-access memory (RAM)

A computer system's short-term memory, used to store calculation data.

read-only memory (ROM)

A place where data that is needed for the computer to function, such as firmware and start-up instructions, is usually stored.

register

A place in a CPU where data currently being used can be stored temporarily.

relational database

The most common type of database, where information is stored in tables. The tables are made up of rows, called records, and columns, called fields.

resolution

A measure of the amount of information in a computer file, such as an image.